

Treeleport

Input file:	standard input
Output file:	standard output
Time limit:	2 seconds
Memory limit:	512 megabytes

You are given a tree consisting of n vertices, numbered from 1 to n . The tree has $n - 1$ bidirectional edges, and traversing any of these edges takes exactly 1 unit of time.

To accelerate travel across the network, m teleportation protocols have been installed. The i -th protocol is described by three integers x_i, u_i , and v_i . This protocol establishes a two-way portal between vertex x_i and each vertex lying on the simple path from u_i to v_i , inclusive. Traveling through any of these portals takes exactly 1 unit of time, regardless of the distance between the two vertices in the original tree, and each portal can be used any number of times.

Your task is to find the minimum total time required to travel from vertex 1 to vertex n .

Input

The first line contains two integers n and m ($1 \leq n, m \leq 10^5$) — the number of vertices in the tree and the number of teleportation protocols, respectively.

Each of the next $n - 1$ lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n, u_i \neq v_i$) denoting a bidirectional edge between vertex u_i and vertex v_i . It is guaranteed that the given edges form a valid tree.

Each of the next m lines contains three integers x_i, u_i , and v_i ($1 \leq x_i, u_i, v_i \leq n$) describing the i -th teleportation protocol.

Output

Print a single integer — the minimum time required to reach vertex n starting from vertex 1.

Example

standard input	standard output
6 2 5 4 2 3 1 4 2 1 5 6 2 5 6 1 3 3	2

Note

The graph of the sample is shown below, with the shortest path highlighted in red.

